

Transaction Concept Summary



give a summary of 17.1, 17.2, 17.3 etc of that chapter in details but explain easily with real life example

Here's a clear and real-life example-based summary of sections 17.1 to 17.4 of *Database System Concepts* (Silberschatz, Korth, Sudarshan):

17.1 Transaction Concept

A **transaction** is a group of operations treated as a single, indivisible unit. Think of it like transferring money from your checking account to your savings account through a banking app. This transfer involves:

- Deducting money from the checking account
- Adding money to the savings account

Both steps **must** succeed together. If your app crashes after deducting the money but before adding it to savings, you'd lose money! That's why databases enforce **ACID** properties:

ACID Properties (with examples):

- **Atomicity** (All-or-nothing): If the bank app crashes after deducting \$50 from account A but before adding it to account B, the system should undo the deduction. Either both happen, or none.
- **Consistency** (No corruption): If accounts A and B together originally had \$3000, they should still have \$3000 after the transaction (even if \$50 moves between them). No money should magically appear or disappear.
- **Isolation** (No interference): Imagine two people transferring money at the same time. Each person should feel like they were the only one doing transactions. Their actions shouldn't mix up and corrupt the results.
- **Durability** (It sticks!): Once the transfer is complete and confirmed, even if the power goes out, the change (money moved) must still be there when the system restarts.

17.2 A Simple Transaction Model

This section simplifies how transactions are handled internally using a mini-language.

Imagine:

```
plaintext

`read(A)
A = A - 50
write(A)
```

```
read(B)
B = B + 50
write(B)
\
```

These steps represent transferring \$50 from A to B. The system must ensure that if it crashes halfway through, it can undo the changes, or if it finishes, the result stays permanent.

Real-Life Analogy:

Think of writing a check:

- You check your balance (read A)
- Subtract the amount ($A = A - 50$)
- Hand it to the bank (write A)
- The bank credits the other account (read B, $B = B + 50$, write B)

If the bank system crashes between write A and write B, only one account changes — that's a problem unless recovery fixes it (which is where atomicity and durability help).

17.3 Storage Structure

Understanding how data is stored helps explain **atomicity and durability**.

Three kinds of storage:

- **Volatile storage:** RAM. Fast, but all data is lost if power fails.
- **Non-volatile storage:** Hard drives, SSDs. Data persists after crashes.
- **Stable storage:** Conceptually never fails (in reality, it's approximated using redundant copies). Think of it like using a cloud backup for your bank transaction logs.

To ensure durability and atomicity, the system stores transaction logs on **stable storage** before actually changing the database. So if anything fails, it can recover from the logs.

17.4 Transaction Atomicity and Durability

This section focuses on **what happens if a transaction fails**:

- A **committed** transaction is one that completed successfully. Its results stay.
- An **aborted** transaction is one that failed. Its effects must be **rolled back**.

To support rollback:

- Every change is recorded in a **log** (before making it).
- If the transaction fails, the log helps undo the changes.
- If it succeeds, the log helps **redo** changes in case of a crash after the transaction but before saving the data.

Real-Life Analogy:

Say you're editing a document:

- You **autosave** versions as you go (like the transaction log).
- If your computer crashes before clicking "Save," your word processor restores the last autosaved version — that's atomicity and durability in action.

Let me know if you want me to continue with the next sections (like 17.5 Isolation) or create a diagram to visualize ACID with these examples!



yes for both

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↓ thanks for your feedback

Sure! Let's now move on to the next section:

17.5 Transaction Isolation

Isolation ensures that **multiple transactions can happen at the same time** without messing each other up.

Real-Life Example: Shopping Carts

Imagine two people are shopping online and trying to buy the last available laptop.

- **User A** adds the laptop to the cart.
- **User B** adds the same laptop a second later.
- Without isolation, both might get confirmation and pay... but only one laptop exists!

Isolation ensures that **only one user successfully completes the transaction**, and the other is informed it's out of stock.

How Databases Do This:

Databases use **scheduling** to interleave actions from different transactions but in a way that the **final result is the same as if they were run one after another**.

That's called a **serializable schedule**.

Types of Problems If Isolation Is Weak:

1. **Dirty Reads:** You see someone's uncommitted data.
 - Example: You check your bank balance and see \$5000, but that was part of a failed transfer that was rolled back.
 2. **Lost Updates:** Two transactions overwrite each other's changes.
 - Example: Two doctors update your hospital record. One updates your medicine, the other updates your allergy info. If not isolated, one change may be lost.
 3. **Temporary Inconsistency:** Midway state is visible.
 - Example: You book a flight, and midway the ticket count temporarily goes negative. Another transaction reading that data may think something is wrong.
 4. **Unrepeatable Read:** You read the same data twice in a transaction and get two different results.
 - Example: You check the price of a stock once at \$100. Before you buy it in the same transaction, the price becomes \$110.
-

Let me know if you want me to continue with **17.6 Serializability** or if you'd like another visual illustration!



can you explain more transaction isolation

Absolutely! Let's dive deeper into **Transaction Isolation**, but explained in a way that feels intuitive — like talking through real-world examples step by step.

What is Transaction Isolation?

Isolation means that **each transaction should run as if it were the only one in the system**, even though many may be running at the same time.

Think of a **busy café**:

- Many baristas are working on orders (transactions) at once.
- Isolation ensures that each drink is made correctly, even if multiple people are ordering drinks at the same time.

Without isolation, one person might get the wrong coffee or two people might get charged for the same latte.

The Four Classic Isolation Problems (with relatable examples):

1. Dirty Read

You see data that was written by another transaction **that hasn't been committed yet**.

 Example:

- Alice starts a transaction and updates her account balance from \$1000 to \$500 (maybe she transferred \$500).
- Bob reads Alice's balance before she finishes and commits.
- Alice then **rolls back** the transaction — her balance is **still** \$1000.
- But Bob made a decision based on **temporary, incorrect data**.

 Why it's bad: You're trusting data that may disappear!

2. Unrepeatable Read

You read the same data twice within a transaction, and it changes between reads — **someone else changed it**.

 Example:

- You check your marks: 75/100.
- You go get a coffee.
- When you check again, it shows 80/100.
- What?! The teacher edited your marks while you were in the middle of checking.

🎯 Why it's bad: You expect consistency inside your own transaction.

3. Lost Update

Two transactions read the same data and both write back, but **only one change survives**.

🧑 Example:

- A nurse updates your hospital room number to Room 210.
- At the same time, an admin updates your emergency contact info — but also edits the room number by mistake (still 205).
- When the admin saves it, your **room number goes back to 205**, losing the nurse's update.

🎯 Why it's bad: Someone's change is lost in the process.

4. Phantom Read

You perform a query like "Give me all students with GPA > 3.5," and then **later in the same transaction**, you do it again... and get **different results** because someone added or deleted data.

🧑 Example:

- You view a list of top 5 students.
- Then someone registers for a course, gets a GPA boost, and suddenly appears in the top 5.
- You refresh — the list changed, even though you're still in your transaction!

🎯 Why it's bad: You thought you had a snapshot of data, but it keeps changing.

📌 Isolation Levels in SQL (from Weak to Strong)

SQL defines **4 levels** of isolation to balance safety vs performance.

Isolation Level	Allows	Blocks	Use Case
Read Uncommitted	Dirty Reads, Unrepeatable Reads, Phantoms	Nothing	Fast, unsafe (rarely used)
Read Committed	✅ Dirty Reads blocked	❌ Unrepeatable Reads allowed	Default in many DBs
Repeatable Read	✅ Dirty + Unrepeatable blocked	❌ Phantom Reads allowed	Safer, but more locking
Serializable	All blocked	Slowest but safest	For super-critical data

🔒 Serializable = Like each transaction is run **one after the other**, no interference at all.

👁 Real-Life Analogy: Google Docs vs Word File

- **Google Docs** (low isolation): Multiple people can edit at the same time, and changes appear live. Great for collaboration, but risk of overwrites.

- **Word File on USB** (high isolation): Only one person edits. Others have to wait. No mix-ups — but slower.

Databases try to find a balance between **speed** and **safety**, depending on your need.

Would you like a chart showing isolation levels vs anomalies? Or want to jump into **17.6 Serializability**?



can you explain isolation levels in sql in more detail

Of course! Let's break down **SQL Isolation Levels** in a **detailed but simple way**, using real-life scenarios and how each level handles the four classic problems: **Dirty Reads**, **Unrepeatable Reads**, **Phantom Reads**, and **Lost Updates**.



What Are SQL Isolation Levels?

SQL lets you control how "isolated" each transaction should be from others. This is done using **isolation levels**.

Think of isolation levels like different levels of privacy in a group chat:

- Some let everyone see your draft messages.
- Others keep your messages hidden until you hit "send."

Each level makes a trade-off:

-  **More isolation** = safer but slower.
-  **Less isolation** = faster but riskier.



The 4 SQL Isolation Levels

Isolation Level	Dirty Read	Unrepeatable Read	Phantom Read	Performance
Read Uncommitted	✅ Possible	✅ Possible	✅ Possible	⚡ Fastest (least safe)
Read Committed	❌ Avoided	✅ Possible	✅ Possible	⚡ Fast, commonly used
Repeatable Read	❌ Avoided	❌ Avoided	✅ Possible	🟡 Slower
Serializable	❌ Avoided	❌ Avoided	❌ Avoided	🔒 Slowest (most safe)

Let's explain each one with examples 🙋



1. Read Uncommitted — "The Gossip Level"

 You can read uncommitted changes made by other transactions (aka **Dirty Reads**).

 Example:

- Bob starts transferring money.

- You read Bob's balance while it's still in-progress and **not committed**.
- Later, Bob's transfer fails, but you already made decisions based on his temporary data.

🎯 Problems:

- Dirty reads ✓
- Unrepeatable reads ✓
- Phantom reads ✓

🚫 Use rarely — usually only for quick reports where **absolute accuracy isn't needed**.

🟢 2. Read Committed — "The Polite Listener"

💬 You only see **committed changes** from others.

📦 Example:

- Bob transfers money.
- You check his balance, but the DB waits until he's done (committed) before showing the new balance.

🎯 Fixes Dirty Reads ❌ But still allows:

- Unrepeatable reads ✓ (You can read Bob's balance twice and get different results.)
- Phantom reads ✓

✅ **Default in many databases** like Oracle, SQL Server.

🟡 3. Repeatable Read — "The Careful Note-Taker"

💬 Once you read a row, **it stays the same** throughout your transaction.

📦 Example:

- You check Alice's grades — 85%.
- Even if someone updates it during your transaction, you **still see 85%** unless you refresh outside.

🎯 Fixes:

- Dirty reads ❌
- Unrepeatable reads ❌

🚫 But not Phantom Reads ✓ Example: If you query `WHERE GPA > 3.5`` and someone inserts a new high-GPA student during your transaction, you'll see it in a second query.

✅ Good for banking, order systems where **row-level accuracy** is critical.

🔴 4. Serializable — "The VIP Room"

💬 It's like transactions run **one at a time**, in a **strict order**. Safest level.

📦 Example:

- You check for available train seats.
- Even if someone else tries to book a seat at the same time, the DB ensures that one transaction finishes before another starts.

🎯 Fixes:

- Dirty reads ❌
- Unrepeatable reads ❌
- Phantom reads ❌

🔴 Slowest — can lead to **more locks, wait times, or even deadlocks**. ✅ Best for **critical systems** (finance, healthcare, etc.)

🔄 How to Set Isolation Levels in SQL

You can set the isolation level using a command like:

```
sql

\SET TRANSACTION ISOLATION LEVEL SERIALIZABLE;
BEGIN TRANSACTION;
-- your queries here
COMMIT;
```

Supported levels may vary a bit:

- **PostgreSQL** supports all levels.
- **MySQL (InnoDB)** supports everything except **Read Uncommitted** by default.
- **SQL Server** and **Oracle** use **Read Committed** as default.

🧠 TL;DR Summary (with Emoji Ratings)

Level	Safe?	Speed	Good For
🔓 Read Uncommitted	❌ Unsafe	⚡ Fastest	Quick logs, diagnostics
🔒 Read Committed	✅ Decent	⚡ Fast	Most general apps
🔒 Repeatable Read	✅ Safer	🟡 Medium	Banking, records
🛡️ Serializable	✅✅ Safest	🔴 Slowest	Payments, high-value data

Would you like a **visual cheat sheet** or help writing SQL examples for each level?



there are more topics in 17.5 in the book which you have missed

You're right! Section **17.5: Transaction Isolation** includes much more than just isolation levels — let me now give you a **full detailed and simple explanation** of **everything covered in 17.5**, based on the book *Database System Concepts* by Silberschatz, Korth, and Sudarshan.

✓ Full Explanation of 17.5 Transaction Isolation

This section is not just about isolation levels — it also explains:

1. Why we need **concurrent execution**
2. **Schedules** (the order of operations in transactions)
3. How to identify **correct schedules**
4. The **types of conflicts and problems** isolation solves

Let's break it down clearly and simply:

1. Why Allow Concurrent Transactions?

Running multiple transactions at once **improves performance** in two major ways:

a. Improved Throughput & Resource Utilization:

- Some transaction steps involve **CPU**, others involve **disk I/O**.
- While one transaction is **reading from disk**, another can **use the CPU**.
- More things happen **in parallel**, so we finish more work in the same time.

b. Reduced Waiting Time:

- If we run things one-by-one (serially), a short task may have to wait for a long one.
- But if they use **different parts of the database**, they can run at the same time.
- Result? Faster average response time.

💡 Think of it like:

“One person boiling pasta and another chopping vegetables — both can work at once, rather than one waiting for the other.”

2. What is a Schedule?

A **schedule** is the **chronological order** in which the operations (read, write, commit) of transactions are executed.

 **Types of schedules:**

- **Serial Schedule:** One transaction finishes before another begins.
- **Concurrent Schedule:** Instructions of multiple transactions are **interleaved**.

Examples:

Let's say:

- **T1:** Transfer \$50 from A to B
- **T2:** Transfer 10% from A to B

These can be arranged in different ways:

- T1 fully, then T2 ⇒ **Serial**
- Interleaved steps of T1 and T2 ⇒ **Concurrent**

🧠 But not all interleavings are safe — they may corrupt data!

💡 3. Why Isolation is Needed: Problem Scenarios

Concurrent execution can cause **problems**:

- One transaction might **read data being changed** by another.
- Final values may depend on **who gets CPU time first**.

Thus, the **Isolation** property says:

“The final result should look as if the transactions ran one at a time in some order.”

🕵️ 4. Identifying Safe Schedules: Serializability

A schedule is **serializable** if it's **equivalent to some serial execution**.

That means:

- Even if interleaved, it gives **correct results**, as if one transaction followed the other.

This is the core goal of **concurrency control**.

⚠️ 5. Types of Conflicts That Break Isolation

When you run transactions together, these three types of **conflicts** can break isolation:

Conflict Type	Meaning	Example
Read-Write (RW)	A transaction reads, another writes	You read \$100 balance, but another transaction updates it to \$80 before you finish
Write-Read (WR)	A transaction writes, another reads	You deduct money, but someone else reads the account while the transaction isn't done
Write-Write (WW)	Both write to the same data	Two people update the same record at the same time

A schedule that avoids these in the wrong order can be **serializable**, and hence safe.

6. Role of Concurrency Control

Concurrency-control mechanisms ensure that:

- Even when transactions are interleaved,
- The final result is **consistent**
- The operations behave as though executed **in some serial order**

These mechanisms are described in **Chapter 18**, but this section gives the **reason and need** for such controls.

Key Takeaways from 17.5

- Concurrent execution is **good** for performance.
 - But it creates **risks** of inconsistency.
 - A **schedule** shows how operations interleave.
 - Some schedules are safe (serializable), others are not.
 - **Isolation** means transactions should behave as if they ran one-by-one.
 - The **database must enforce isolation** through internal mechanisms.
-

Would you like me to:

- Summarize **17.6 Serializability** next?
- Or give you a **diagram or example** showing serial vs non-serial schedules?

Let me know!



can you explain these with example,please

Absolutely! Let's walk through **everything from Section 17.5** with **clear, relatable examples**, especially focusing on:

1. Why we allow concurrent transactions
 2. What a schedule is
 3. When and why schedules can go wrong
 4. What **serializability** means
 5. Types of **conflicts** that violate isolation
-

1. Why Allow Concurrent Transactions?

Real-Life Analogy: A Kitchen

- **Chef A** is boiling pasta (takes 10 mins).
- **Chef B** is chopping vegetables (takes 5 mins).
- If B waits for A to finish boiling before chopping — that's **serial execution** — total 15 mins.
- If both work **at the same time**, total time is **just 10 mins**.

 Similarly in databases:

- Transaction T1 may be waiting for disk.
- Meanwhile, T2 can use CPU.
- Running them **in parallel saves time**.

2. What is a Schedule?

A **schedule** is a timeline of steps taken by different transactions.

Example: Two Transactions

Let's say:

- T1: Transfer \$100 from A to B
- T2: Apply 10% interest to A

Initial values:

- A = 1000
- B = 500

Serial Schedule (T1 then T2)

CSS

```
T1: Read(A) → A = A - 100 → Write(A)
T1: Read(B) → B = B + 100 → Write(B)
T2: Read(A) → A = A × 1.10 → Write(A)
```

Final values:

- $A = (1000 - 100) \times 1.10 = 990$
- B = 600

 Safe

Interleaved (Concurrent) Schedule:

CSS

```

T1: Read(A) → A = A - 100
T2: Read(A) → A = A × 1.10
T1: Write(A)
T2: Write(A)

```

Let's compute:

- T1 reads $A = 1000 \rightarrow A = 900$
- T2 reads $A = 1000 \rightarrow A = 1100$
- T1 writes $A = 900$
- T2 writes $A = 1100$ 🤖 (T1's change is lost!)

Final result:

- $A = 1100$
- $B = \text{unchanged}$

✗ **Incorrect** result — this schedule is **not serializable**

✂ 3. Types of Conflicts (with Examples)

Let's define 3 types of **conflicting operations** that cause unsafe interleaving.

a. Read-Write (RW) Conflict

- T1 reads A
- T2 writes A

🔄 Example:

- T1 checks account balance
- T2 updates the balance mid-way
- T1 uses outdated data

b. Write-Read (WR) Conflict

- T1 writes A
- T2 reads A

🔄 Example:

- T1 deducts \$100 from A
- Before T1 finishes, T2 reads the new (unconfirmed) value of A

Result? T2 uses **uncommitted** (dirty) data.

c. Write-Write (WW) Conflict

- T1 writes A
- T2 also writes A

 Example:

- Two transactions both update a product's stock
- The last one to write **overwrites** the other

4. What is Serializability?

A schedule is **serializable** if it gives the **same result** as if the transactions were run one after another (serially).

- ✓ Serializable = Isolated, consistent result
- ✗ Not Serializable = Potential corruption or logical errors

•• Final Example: Three Schedules

Let's say T1: deposit \$100 to A

T2: apply 10% interest to A

Initial A = \$1000

A. Serial (T1 then T2):

```
yaml
`T1: A = A + 100 → A = 1100
T2: A = A × 1.1 → A = 1210
✓ Result: A = 1210`
```

B. Serial (T2 then T1):

```
yaml
`T2: A = A × 1.1 → A = 1100
T1: A = A + 100 → A = 1200
✓ Result: A = 1200`
```

Both valid (just different order).

C. Non-Serializable Interleaved:

```
yaml
`T1: Read A = 1000 → A = 1100
T2: Read A = 1000 → A = 1100`
```

```
T1: Write A = 1100
T2: Write A = 1100
✗ Final A = 1100 (should be 1210 or 1200)
```

☀ Here, **T1's and T2's updates overwrite each other**. One is lost.

Summary

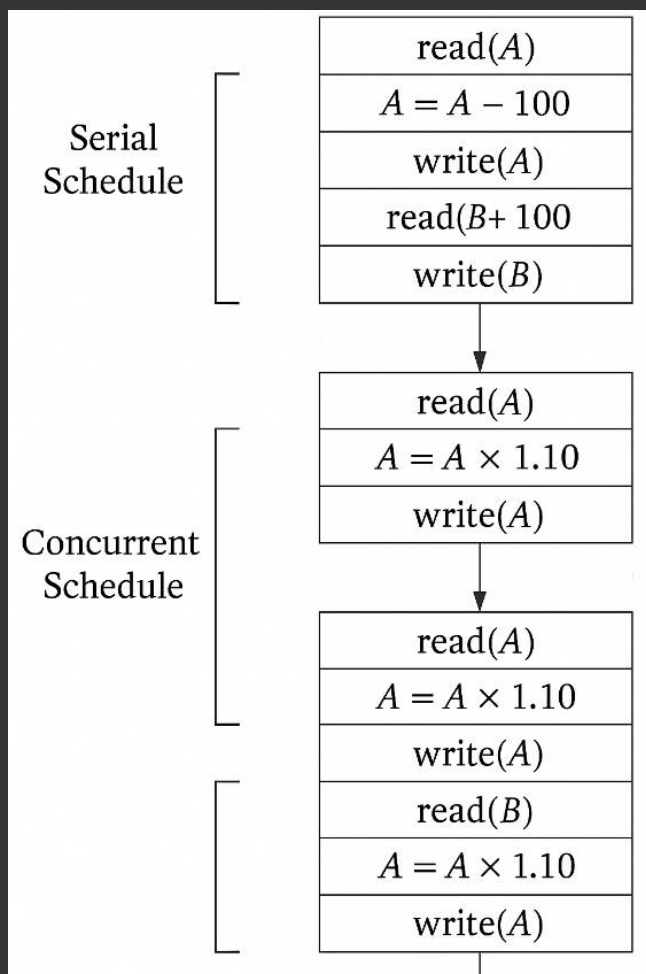
Concept	Example	Key Insight
Concurrent Execution	Two chefs working at once	Improves speed
Schedule	Timeline of read/write steps	Can be safe or unsafe
Conflict	Two transactions read/write same data at the same time	Risk of errors
Serializable	Same result as one-by-one execution	Goal of isolation
Isolation Needed	To avoid lost updates, dirty reads, inconsistencies	Ensures data safety

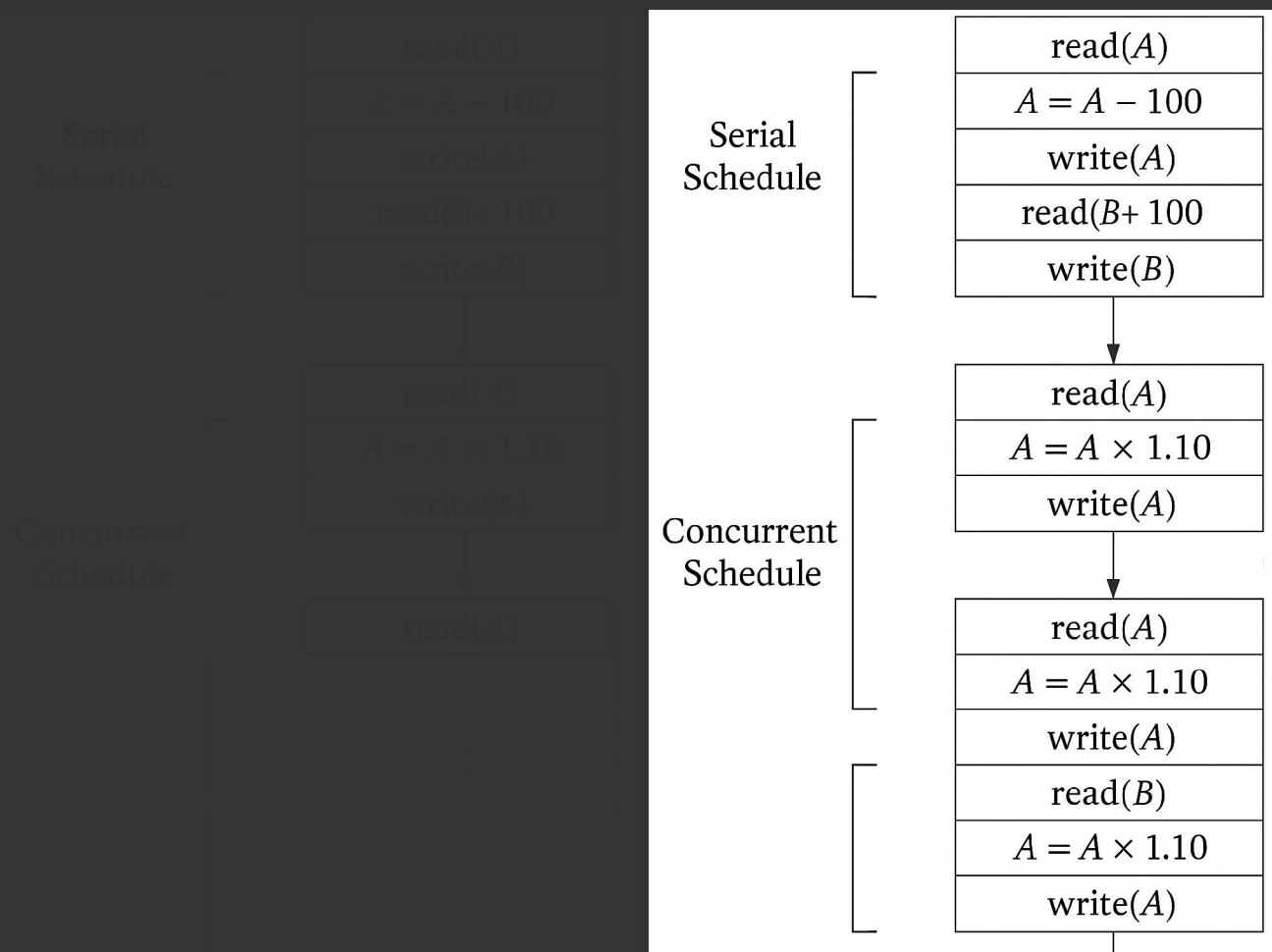
Would you like a **visual timeline** of the serial vs interleaved schedules? I can generate that next!

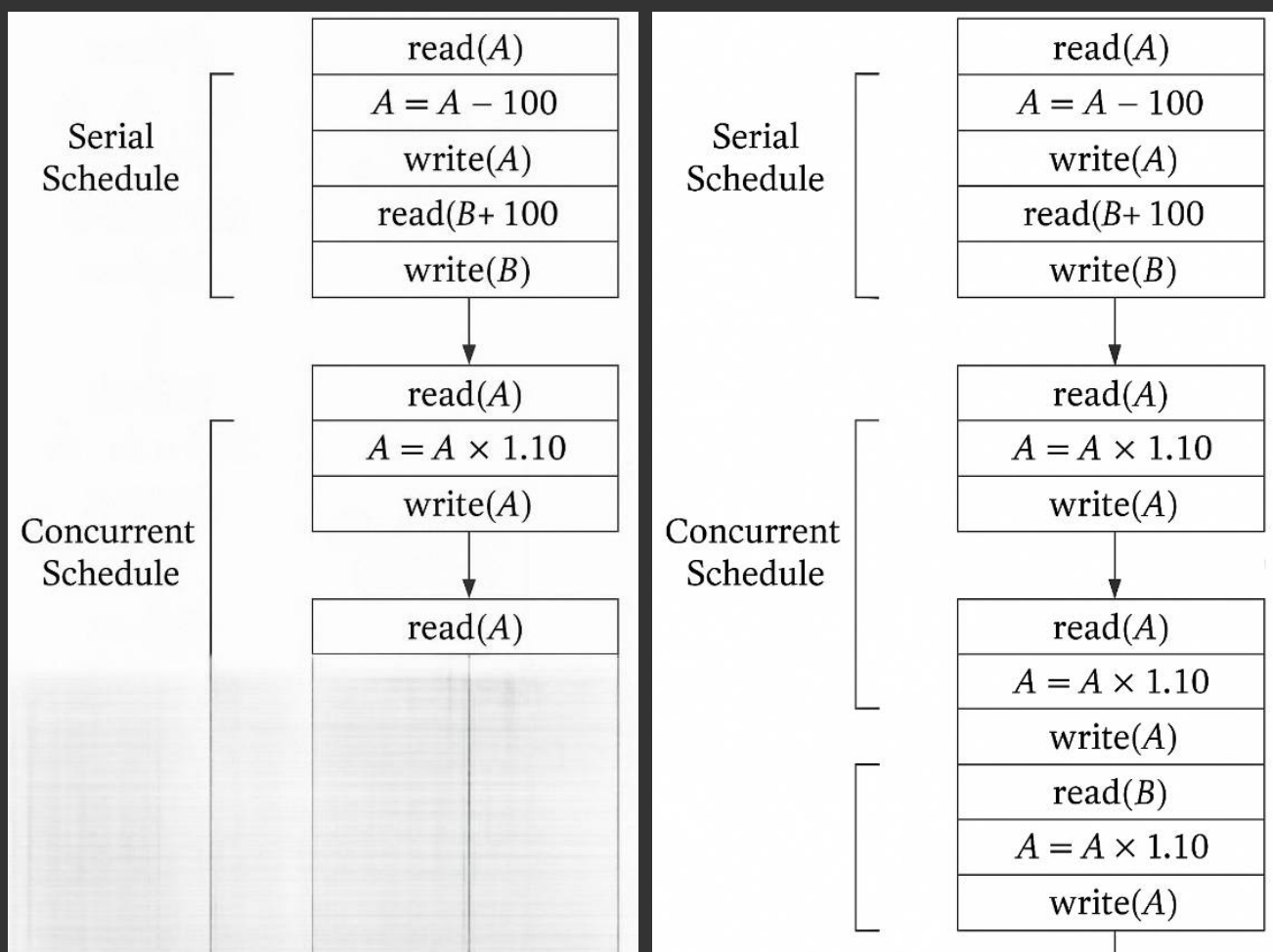


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 Like this image

Here you go! Let me know if you'd like the same for conflict types or isolation levels.



now explain the next topic from the book, please dont miss any topic

Here's a detailed and simple explanation of **Section 17.6: Serializability** from the *Database System Concepts* book, with clear real-world examples and diagrams where useful.

What is Serializability?

When many transactions run together (concurrently), we want the **final result** to be **correct** — as if the transactions had run one after another.

That's **serializability**:

A schedule is **serializable** if it gives the same result as some **serial execution**.

Why is This Needed?

Problem Example (Inconsistent Schedule):

Two transactions:

- **T1**: Withdraws \$50 from account A and deposits into B.
- **T2**: Applies 10% interest to account A and B.

If these two run **without careful control**, the database might:

- Add \$50
- Or apply interest to an already-decreased balance
- Or lose money altogether

This leads to **inconsistent state**.

Schedule: Serial vs Interleaved

- **Serial Schedule**: One transaction fully completes before another starts.
- **Concurrent Schedule**: Instructions from both transactions are mixed.

Not all interleaved schedules are **safe** — we need a way to **check** if an interleaved schedule is still **serializable**.

Conflict Serializability (Main Focus of 17.6)

To simplify, the book focuses on only **read** and **write** operations.

Conflicting Operations:

Two actions **conflict** if:

- They access the **same data item**, and
- At least one is a **write**

Operation 1	Operation 2	Conflict?
read(A)	read(A)	✗ No
read(A)	write(A)	✓ Yes
write(A)	read(A)	✓ Yes
write(A)	write(A)	✓ Yes

Swapping Non-Conflicting Steps

If two steps don't conflict, we can **swap** them without changing the outcome.

Example:

```
bash
```

```
`T1: read(A)
T2: read(B)
`
```

No conflict → safe to swap.

If we can rearrange the schedule by **swapping non-conflicting instructions** into a serial form, the schedule is **conflict serializable**.

Precedence Graph (Dependency Graph)

To test if a schedule is **conflict serializable**, we draw a **precedence graph**:

- **Nodes:** Each transaction
- **Edges:** $T_i \rightarrow T_j$ if T_i does something **before** T_j that conflicts

Conflict rules:

- T_i writes(Q) **before** T_j reads(Q)
- T_i reads(Q) **before** T_j writes(Q)
- T_i writes(Q) **before** T_j writes(Q)

 **If the graph has a cycle → ✗ Not serializable**

If **no cycle** →  **Serializable**

Example: Detecting Serializability via Precedence Graph

Schedule:

```
css
`T1: read(A), write(A)
T2: read(A), read(B), write(B)
`
```

Conflicts:

- T1's `write(A)` before T2's `read(A)` → edge: **T1** → **T2**

Graph:

```
nginx
`T1 → T2
`
```

✓ No cycle → Conflict serializable

Another Example: Non-Serializable

```
css
`T1: read(A), write(B)
T2: write(A), read(B)
`
```

Conflicts:

- T2's `write(A)` before T1's `read(A)` → **T2** → **T1**
- T1's `write(B)` before T2's `read(B)` → **T1** → **T2**

Graph:

```
nginx
`T1 → T2
T2 → T1
`
```

✗ Cycle found → Not serializable

Topological Sorting (Extra Insight)

If the precedence graph has no cycles, you can do a **topological sort** to find the **serial order** of transactions.

Example:

```
nginx
```

```
`T3 → T2
T2 → T1
`
```

Possible serial orders: $T3 \rightarrow T2 \rightarrow T1$

⚠ Conflict Serializability vs View Serializability

- **Conflict serializability** is easier to check (so it's used in practice).
- **View serializability** allows more schedules (less strict), but is harder to test (NP-complete!).

🧠 Summary of Key Concepts in 17.6

Concept	Meaning
Schedule	Order of operations from multiple transactions
Conflict	When operations access same data and at least one is a write
Conflict Serializability	Can be converted into a serial schedule by swapping non-conflicting steps
Precedence Graph	Graph showing which transaction depends on which
Cycle in Graph	Means the schedule is not conflict serializable

Would you like a visual graph-based explanation of one of the examples from this section? I can generate a diagram of a **precedence graph with and without a cycle** to show this!



yes

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Here you go! Let me know when you're ready to continue with Section 17.7 or need a breakdown of view serializability too.



next

Here's a detailed and simple explanation of **Section 17.7: Transaction Isolation and Atomicity** from *Database System Concepts* (Silberschatz, Korth, Sudarshan), with real-life examples for better understanding.

🧩 Why 17.7 Matters

So far, we've assumed that transactions don't fail. But what if one fails *in the middle* of execution, and others have already **read or acted on its changes**?

This section addresses the risks and rules of maintaining **atomicity** (all or nothing) and **isolation** in **concurrent transaction failures**.

🔥 Problem: What if a Transaction Fails After Another Has Seen Its Changes?

Let's say:

- **T6** writes to A
- **T7** reads A (updated by T6)
- Then T7 commits
- But **T6 later fails!**

Now T7 has committed based on **invalid data** — this **breaks atomicity**, because we can't roll back T7 anymore.

🚫 This is a **non-recoverable schedule**.

✅ 17.7.1 Recoverable Schedules

A **recoverable schedule** is one where:

A transaction that **reads data from another transaction** only **commits after** the first one commits.

👤 Real-life Example:

Imagine two doctors updating a patient's treatment:

- Doctor A updates the prescription (T6).
- Doctor B (T7) reads that new prescription and commits it to the hospital system.
- But then Doctor A's transaction fails.

 To be safe, Doctor B should **wait until Doctor A's update is confirmed**.

Bad (Non-Recoverable) Schedule

```
makefile
`T6: write(A)
T7: read(A)
T7: commit
T6: crash (never commits)
`
```

 T7 should not have committed!

Good (Recoverable) Schedule

```
makefile
`T6: write(A)
T7: read(A)
T6: commit
T7: commit
`
```

 T7 waits for T6's commit — now we're safe.

17.7.2 Cascadeless Schedules

Even if we make schedules **recoverable**, we might still face a **cascading rollback** if one transaction fails.

Example: Chain of Dependencies

```
CSS
`T8: write(A)
T9: read(A) → write(A)
T10: read(A)
`
```

If **T8 fails**, then:

- T9 must be aborted (it read T8's value)
- Then T10 must also be aborted (it read T9's value)

 This is called **cascading rollback**.

Cascadeless Schedule Rule:

A transaction must **only read committed data** (no dirty reads).

That means: **T9 should not read from T8 until T8 commits.**

Key Takeaways from 17.7

Concept	What It Means	Why It Matters
Recoverable Schedule	Don't commit after reading uncommitted data	Ensures rollback is safe
Cascading Rollback	Chain of aborts caused by one failed transaction	Bad for performance
Cascadeless Schedule	Transactions only read committed data	Avoids cascading failures

Visual Concept (Timeline View)

Let me know if you'd like a diagram that shows:

- A **non-recoverable** schedule
- A **recoverable** one
- A **cascading rollback**
- A **cascadeless** schedule

We can walk through it visually!

Would you like that now?



next topic

Here's a clear and example-driven explanation of **Section 17.8: Transaction Isolation Levels** from *Database System Concepts* by Silberschatz, Korth, and Sudarshan.

17.8 Transaction Isolation Levels

We've already seen that **serializability** is the ideal — it ensures correctness when multiple transactions run at the same time. But in real life, perfect isolation can be **too slow** or **block too many users**, especially for long-running queries.

So, SQL lets you **relax isolation** for better performance — at the cost of **more responsibility on the programmer** to avoid bugs.

The Four SQL Isolation Levels

Level	Dirty Reads	Unrepeatable Reads	Phantom Reads	Performance	Used In
Read Uncommitted	✅ Allowed	✅ Allowed	✅ Allowed	⚡ Fastest	Logs, diagnostics
Read Committed	❌ No	✅ Allowed	✅ Allowed	⚡ Fast	Default in many DBs
Repeatable Read	❌ No	❌ No	✅ Allowed	🟡 Moderate	Financial apps
Serializable	❌ No	❌ No	❌ No	🔴 Slowest	Critical systems

What These Mean (With Examples)

1. Read Uncommitted — “You can see everyone’s scribbles”

- You can read values that were changed by **uncommitted transactions** (aka **dirty reads**).
- Example:
 - Alice transfers money but doesn’t commit.
 - Bob sees the updated balance.
 - Then Alice’s transaction fails. Now Bob’s read was invalid!

⚠️ **Highly unsafe**, rarely used.

2. Read Committed — “You can only see published edits”

- You can only read data that is **committed**.
- But someone can still **change the value between two reads** → **unrepeatable read**.

 Example:

- You check your train ticket price at 10:00 AM (\$100).
- You check again at 10:01 AM — now it’s \$120.
- Another transaction updated it and committed between your reads.

✅ Safer than before, still common as default.

3. Repeatable Read — “You get a snapshot of rows you read”

- If you read something once, **no one can change it** until your transaction ends.
- However, **new rows** satisfying your conditions can still appear = **phantom reads**.

 Example:

- You run: ``SELECT * FROM orders WHERE price > 100``.
- You see 10 rows.
- Another transaction inserts a new \$150 order and commits.

- You run the query again — now you see 11 rows.

● Still not perfect — but solid for many apps.

4. Serializable — “Everyone works in line — one at a time”

- It blocks **all anomalies**: no dirty reads, no repeat issues, no phantom records.
- Every transaction behaves as if **it ran alone**.

 Example:

- If you check how many seats are left, and someone else tries to book, they’ll wait until your transaction finishes.
- Most strict — and safest.

 Used in critical systems like banking.

Bonus Concept: Dirty Writes (Disallowed by All)

Even the weakest isolation level **blocks dirty writes**:

No transaction can **write to a data item** if another transaction **has already written to it** and hasn’t committed.

This prevents mutual overwriting and total chaos!

How to Set Isolation Level in SQL

SQL Syntax (Example for Serializable):

```
sql
`SET TRANSACTION ISOLATION LEVEL SERIALIZABLE;
BEGIN;
-- your SQL here
COMMIT;
`
```

In JDBC:

```
java
`connection.setTransactionIsolation(Connection.TRANSACTION_SERIALIZABLE);
`
```

Summary Table with Real-World Analogy

Level	Analogy	Risk of Errors
Read Uncommitted	You read your friend's unsaved essay	🔥 Very high
Read Committed	You read only saved versions	⚠️ Moderate
Repeatable Read	You get your own consistent copy	✅ Safer
Serializable	You use the library when no one else is allowed in	🔒 Safest

Would you like a diagram showing the different anomalies (dirty read, unrepeatable read, phantom) with each isolation level?



next topic

Here's a clear and example-based explanation of **Section 17.9: Implementation of Isolation Levels** from *Database System Concepts* (Silberschatz, Korth, Sudarshan).

🔧 17.9 Implementation of Isolation Levels

This section explains **how databases actually implement isolation**, using various concurrency control techniques. Just *declaring* an isolation level isn't enough — **how it's enforced behind the scenes** matters a lot for correctness and performance.

🔒 17.9.1 Locking (Most Common)

Locking is the most straightforward method. Here, a transaction **locks** data items before accessing them.

- **Shared Lock (S)**: For **reading** data.
- **Exclusive Lock (X)**: For **writing** data.

🔄 Real-life Example:

At a library:

- If you're **reading** a book, others can too (shared lock).
- If you're **editing** the book, you need to be alone (exclusive lock).

⚙️ Locking Rules:

- You can hold a **shared lock** only if no one has an **exclusive lock**.
- You can hold an **exclusive lock** only if **no one else** has *any* lock.

🔒 Using **two-phase locking** (covered in Ch.18), serializability can be guaranteed.

⌚ 17.9.2 Timestamps

Each transaction gets a **timestamp** when it starts.

The system tracks two timestamps per data item:

- **Read timestamp:** latest transaction that read it
- **Write timestamp:** latest transaction that wrote it

How It Works:

- If a new transaction tries to **access data "out of order"**, it gets aborted and restarted with a new timestamp.

 **Goal:** Transactions execute **in timestamp order**, even if interleaved.

Example:

- T1 (timestamp 5) wants to write A.
- T2 (timestamp 10) already read A.
- Now T1 must abort because it's **too old** to overwrite something T2 read.

17.9.3 Multiversion and Snapshot Isolation

This method keeps **multiple versions** of data items.

- Read-only transactions get a **snapshot** of the database at the time they start.
- Update transactions work on their own version until commit.

Snapshot Isolation:

Each transaction sees a consistent snapshot of the database, unaffected by other concurrent transactions.

Benefit:

- **No locks needed for reads**
- Readers don't block writers, and vice versa → **better performance**

Downside:

- Can cause **non-serializable executions** (e.g., "write skew")

Snapshot Isolation in Real Life

Imagine:

- You take a screenshot of a webpage (your snapshot).
- While you scroll through it, the live site is being updated.

- You **only see what was there when you took the screenshot**.

This isolation lets **long read-only queries run fast**, but updates still have to be checked before committing.

Example of Snapshot Isolation Issue

Let's say:

- T1 reads doctors A and B and decides "At least one is on call."
- T2 reads same doctors and thinks the same.
- Both remove *themselves* from call list.

✗ Neither sees the other's removal → now **no one** is on call! This wouldn't happen in a serializable schedule.

Summary of Isolation Implementation Techniques

Method	Key Idea	Pros	Cons
Locking	Transactions lock data items	Simple, common	May cause blocking, deadlocks
Timestamps	Each transaction has a unique time label	No locks needed	More aborts possible
Multiversion/Snapshot	Each transaction gets a consistent snapshot	Fast reads, high concurrency	May break serializability

Would you like a visual chart comparing these three techniques next, or shall we move on to **Section 17.10: Transactions as SQL Statements**?



17.10

Here's a clear and simple explanation of **Section 17.10: Transactions as SQL Statements** from *Database System Concepts* by Silberschatz, Korth, and Sudarshan.

What This Section Explains

Until now, we talked about **transactions using a simplified model** — like "read A, write B," etc. But in real life, **transactions are written using SQL statements**, like:

```
sql
\UPDATE account SET balance = balance - 100 WHERE id = 'A';
```

This section explores how SQL transactions deal with:

- Inserts
- Deletes
- Complex queries
- Conflicts due to **predicates**
- The **phantom problem**

SQL Transactions in Action

A **SQL transaction** is a group of statements like:

```
sql
`BEGIN;
UPDATE account SET balance = balance - 100 WHERE id = 'A';
UPDATE account SET balance = balance + 100 WHERE id = 'B';
COMMIT;
`
```

Just like in the simple model, this whole set must follow ACID:

- **Atomic** (all or nothing)
- **Consistent**
- **Isolated**
- **Durable**

The Phantom Problem (with Example)

Imagine This SQL Query:

```
sql
`SELECT ID, name FROM instructor WHERE salary > 90000;
`
```

It finds instructors earning over 90K — say “Einstein” and “Brandt.”

Now at the same time, someone else executes:

```
sql
`INSERT INTO instructor VALUES ('11111', 'James', 'Marketing', 100000);
`
```

What Happens?

If the ``INSERT`` happens before your query → you'll see “James.”

If your query happens before the ``INSERT`` → you won't.

👉 This is a **conflict**, even though the insert wasn't to any specific row being read — it's a conflict on a **range** or **predicate**: ``salary > 90000``.

This is the **phantom phenomenon** — the inserted record is a **phantom**: it appears or disappears depending on the schedule.

🔧 Why This Breaks the Simple Model

In the simple model, you track conflicts by looking at **which data item (like A or B)** is read or written.

But in SQL:

- Queries may access **sets** of tuples based on **conditions**.
- You don't always know **which specific rows** are affected just by looking at the SQL.

Example:

```
sql
`UPDATE instructor SET salary = salary * 0.9 WHERE name = 'Wu';`
```

Whether this conflicts with a SELECT depends on:

- Whether Wu's data is **accessed by the query**
- Whether an **index is used** to skip Wu's row

So two identical-looking SQL queries might behave differently!

🧱 Predicate Locking

To handle phantoms and predicate-based conflicts, databases need:

- **Predicate locking**: lock not only on actual data, but also on **conditions** like ``salary > 90000``
- Or: use **index locking** on search keys affected by the WHERE clause

These prevent issues caused by inserts, deletes, and updates that affect what a query **would have seen**.

🧠 Key Points from 17.10

Concept	Explanation
SQL Transactions	Are groups of <code>`INSERT`</code> , <code>`DELETE`</code> , <code>`UPDATE`</code> , <code>`SELECT`</code> , etc.
Phantom Problem	Caused by changes to <i>data ranges</i> (like new rows matching a WHERE clause)
Predicate Conflicts	Occur when conditions (not just individual rows) overlap between transactions
Locking Challenges	SQL queries need advanced locking (e.g., predicate/index locking) to ensure isolation

Would you like a visual illustration of the phantom problem and predicate locking next? Or shall we move on to **Section 17.11: Summary**?



solve the exercises which mainly focuses on transactions, and how to handle conflicts

Great! Let's now go through and **solve the relevant exercises from Chapter 17** of *Database System Concepts*, especially focusing on those that deal with **transactions and conflicts**.

✓ Exercise 17.6 — Conflict Serializability Check

Q: Consider the precedence graph in Figure 17.16. Is the corresponding schedule conflict serializable?

 The figure shows this precedence graph:

SCSS

$T1 \rightarrow T4 \rightarrow T5 \rightarrow T3 \rightarrow T2 \rightarrow T1$ (cycle)

 This forms a **cycle**, so the **schedule is not conflict serializable**.

✓ **Answer:** No, it is not conflict serializable because the precedence graph contains a cycle.

✓ Exercise 17.7 — Cascadeless Schedules

Q: What is a cascadeless schedule? Why is it desirable? Any case where non-cascadeless is okay?

✓ **Answer:**

- A **cascadeless schedule** is one where:

Transactions **only read committed data**, so one failure doesn't force others to roll back.

- **Why desirable?**

It avoids **cascading rollbacks**, saving time and resources.

- **When might non-cascadeless be okay?**

In systems where performance is more critical than consistency (e.g., some analytics systems) and occasional rollbacks are acceptable.

✓ Exercise 17.8 — Lost Update Anomaly

a. Give a schedule showing the lost update anomaly.

 Example:

vbnet

```
T1: read(A) → A = A + 10
T2: read(A) → A = A + 20
T1: write(A) → A = 110
T2: write(A) → A = 120 (T1's change is lost)
```

Initial A = 100 → T1 thinks it's 110, T2 writes 120 → final value ignores T1.

b. Show it's possible with Read Committed isolation.

Yes — Read Committed allows:

- Reading **only committed** data
- But **no guarantee** that data doesn't change between reads and writes

So the above example **can still happen** in this isolation level.

c. Why not possible with Repeatable Read?

Because:

- Under **Repeatable Read**, once a transaction reads a value, **no one can write to it** until the transaction finishes.
- So T2 wouldn't be able to write after T1 reads.

✓ Lost updates are prevented.

✓ Exercise 17.9 — Snapshot Isolation: Banking Example

Describe a nonserializable execution using snapshot isolation in banking.

 Example:

- Checking = \$100, Savings = \$200
- Constraint: Sum ≥ 0
- T36 checks both → thinks total = \$300 → withdraws \$200 from checking
- T37 checks both → same snapshot → withdraws \$200 from savings

Each passes validation — both commit

✗ Final: Checking = -100, Savings = 0 → **constraint violated**

✓ Exercise 17.10 — Snapshot Isolation: Airline Example

Describe a nonserializable execution the airline may tolerate.

 Example:

- 2 passengers check seat availability at same time

- Both pick the last available seat
- Each thinks it's still free and confirms
- Snapshot isolation allows both to proceed

⚠ Conflict arises → system may resolve by:

- Rolling one back
- Or allowing double-booking and resolving later

🎯 Since it's rare and non-critical, the airline might **accept this glitch for better speed.**

Would you like me to solve more exercises from Chapter 18 next (e.g., 18.1 on two-phase locking or timestamp conflicts)?